

# Yacoub Hanna

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## EDUCATION

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### Ph.D. in Computer & Electrical Engineering

August 2026

*Florida International University, Miami, FL*

- GPA: 3.74 / 4.0.

### M.S. in Computer Engineering

August 2025

*Florida International University, Miami, FL*

- GPA: 3.74 / 4.0.

### B.S. in Computer Engineering

December 2020

*Florida International University, Miami, FL*

- GPA: 3.76 / 4.0.

### Associate's degree Candidate

August 2018

*Miami Dade Community College (MDC), Kendall, Florida*

- GPA: 3.95 / 4.0.

### English Proficiency Degree, 5 levels of English language

May 2016

*University of Miami Intensive English Program, Coral Gables, Florida*

### High School Diploma, Electronics Major

June 2014

*Salesian Technical School, Bethlehem, Palestine*

## SKILLS

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- **Cloud, Operating Systems and Virtualization:** Linux (Kali, Ubuntu) | Windows | Raspberry Pi | Virtual Machine Manager | Virtual Box | Google Cloud Platform.
- **Network and Security:** Wireshark | Tcpdump | Open Quantum safe | NS3 | OpenVPN | Stunnel | Bind9 | DNSSEC | DNS over TLS | OpenSSL | Scapy | GNS3 | Open5GS | UERANSIM |
- **Other tools and Technologies:** Docker | Raspberry Pi | Arduino | FTK Imager | Lucidchart | Microsoft Office: Word, Excel, PowerPoint.
- **Programming Languages:** Python | C | C++.

## EMPLOYMENT

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### Florida Gulf Coast University, Fort Myers, FL

August 2026 – Present

#### Visiting Assistant Professor

- Teach undergraduate courses in Computer Security and Digital Systems & Architecture.
- Develop course materials, lectures, laboratory activities, and assessments.
- Conduct research in cybersecurity, Post-Quantum Cryptography (PQC), Internet of Things (IoT), and constrained wireless networks.

### Florida International University, Miami, FL

January 2021 – July 2026

#### Graduate Research Assistant (GA)

- Gained proficiency in Linux command line, troubleshooting, and network setup, facilitating smooth operation in a Linux-based environment.
- Developed and proposed an efficient group key management for resilient operation of smart grid applications, enabling more scalable and secure group communications in resource-constrained IoT.
- Implemented a realistic end-to-end performance evaluation of PQC mechanisms in consumer IoT devices, focusing on constrained resources, wireless connectivity, and secure TLS communication.
- Exhibiting strong teamwork and leadership skills by implementing project management, sharing insights, and resolving issues with diverse team members to ensure the completion of projects and adherence to deadlines.

## WORK EXPERIENCE

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**Florida International University, Miami, FL**

**June 2024 – June 2025**

**Teaching | Foundations of Cybersecurity & Networking Courses**

**Veterans and First Responders Training Initiative (VTI)**

- Providing the foundational knowledge to transitioning military personnel, veterans, and first responders seeking to enter the cybersecurity workforce or advance in their respective careers.

**Skopje, North Macedonia**

**January 2024 – April 2024**

**Teaching Assistant (Hybrid) | Digital Forensics Course**

- Providing help for in-person workshop. Teach them how to access deleted data from various hard drives and how to use HxD, AccessData, FTK Imager, and EPS Viewer.

**Skopje, North Macedonia**

**September 2023 – December 2023**

**Teaching (Hybrid) | Cyber-Physical Systems & Internet of Things Courses**

- Providing Lectures and labs for a duration of 6 weeks for each course. Teaching the fundamentals and the importance of security for each course. Implementing several protocols that are associated with each system, such as MQTT, IEC 61850 Goose, Modbus, DNP3, and the security aspects of each system.

**Florida International University, Miami, FL**

**May (2023 – 2025)**

**Teaching Assistant (In-Person) | ACCESS Step Workshop**

- Gave a workshop to undergraduate and High School students on Cybersecurity and Ethical Hacking. Used Kali Linux, Metasploitable 2, and several tools to perform an enumeration of different systems.

**Ukraine**

**May 2021 – December 2022**

**Teaching Assistant (Online) | Cyber-Physical Systems & Internet of Things Courses**

- Providing labs for each lecture for a duration of 6 weeks for each course. Implementing several protocols that are associated with each system, such as MQTT, IEC 61850 Goose, Modbus, DNP3, as well as security aspects of each system.

## PUBLICATIONS

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- **Y. Hanna**, M. Veksler, and K. Akkaya, (2026). PQ-CKEM: Efficient Quantum resistant Group Key Creation for Large-Scale LEO Satellite Networks. Proceedings of the IEEE International Symposium on Local and Metropolitan Area Networks (LANMAN).
- **Y. Hanna**, S. Tonyali, R. Harrilal-Parchment, and K. Akkaya, (2026). Rethinking IoT Security in the Era of Quantum Computing: Literature Reviews. In Quantum Computing, Sensing and Communications for IoT (pp. 169-226). Singapore: Springer Nature Singapore.
- **Y. Hanna**, J. Bozhko, S. Tonyali, R. Harrilal-Parchment, M. Cebe, and K. Akkaya, "A comprehensive and realistic performance evaluation of post-quantum security for consumer IoT devices." Elsevier Internet of Things Journal 2025, 101650, doi: <https://doi.org/10.1016/j.iot.2025.101650>.
- M. Paudel, **Y. Hanna**, M. Veksler, K. Akkaya, D. Pineda, M. Anastasova, and R. Azarderakhsh, Practical and Efficient Post-Quantum Security Framework for IoT-based 5G Networks. Available at SSRN 6714859.
- **Y. Hanna**, M. Cebe, J. Leon, and K. Akkaya, "Efficient Group Key Management for Resilient Operation of LoRaWAN-Based Smart Grid Applications." IEEE Transactions on Control Systems Technology 32, no. 5, 2024, pp. 1706-1717, doi: 10.1109/TCST.2024.3378988.
- **Y. Hanna**, D. Pineda, M. Veksler, M. Paudel, K. Akkaya, M. Anastasova, and R. Azarderakhsh, "Integrating Post-Quantum TLS into the Control Plane of 5G Networks," 2024 IEEE International Performance, Computing, and Communications Conference (IPCCC), Orlando, FL, USA, 2024, pp. 1-8, doi: 10.1109/IPCCC59868.2024.10850437.
- A. Mutlugun, **Y. Hanna**, and K. Akkaya, "Performance Evaluation of Quantum-Resistant IKEv2 Protocol for Satellite Networking Environments." 2024 IEEE Virtual Conference on Communications (VCC), NY, USA, pp. 1-7, 2024, doi: 10.1109/VCC63113.2024.10914395.
- **Y. Hanna**, D. Pineda, K. Akkaya, A. Aydeger, R. Harrilal-Parchment, and H. Albalawi, "Performance Evaluation of Secure and Privacy-preserving DNS at the 5G Edge," 2023 IEEE 20th International Conference on Mobile Ad Hoc and Smart Systems (MASS), Toronto, ON, Canada, 2023, pp. 89-97, doi: 10.1109/MASS58611.2023.00019.
- J. Bozhko, **Y. Hanna**, R. Harrilal-Parchment, S. Tonyali, and K. Akkaya. "Performance evaluation of quantum-resistant TLS for consumer IoT devices." 2023 IEEE 20th Consumer Communications & Networking Conference (CCNC), Las Vegas, NV, USA, pp. 230-235, 2023, doi: 10.1109/CCNC51644.2023.10060762

- J. Leon, **Y. Hanna**, and K. Akkaya. "Integration of WAVE and OpenFlow for Realization of SDN-based VANETs in ns-3." 2022 Workshop on ns-3 (WNS3), pp. 41-48. 2022, doi: doi.org/10.1145/3532577.3532608
- J. Leon, **Y. Hanna**, and K. Akkaya. "Development and evaluation of a publish/subscribe IoT data sharing model with LoRaWAN." 2022 Open Journal of Internet of Things (OJIOT) 8, no. 1, 2022, pp. 7-19, urn: urn:nbn:de:101:1-2022090515501014226277
- **Y. Hanna**, M. Cebe, S. Mercan, and K. Akkaya, "Efficient group-key management for low-bandwidth smart grid networks." 2021 IEEE International Conference on Communications, Control, and Computing Technologies for Smart Grids (SmartGridComm), Aachen, Germany, 2021, pp. 188-193, doi: 10.1109/SmartGridComm51999.2021.9631988.

## RESEARCH IMPACT METRICS

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### Google Scholar:

- **Citations:** 127
- **H-index:** 7
- **I10-index:** 5
- **Profile:** <https://scholar.google.com/citations?user=jbXCRpEAAAAJ&hl=en&oi=ao>

### ORCID:

- **ORCID ID:** <https://orcid.org/0000-0002-8954-9336>

## RESEARCH FUNDING AND SUPPORT

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- **Cyberskills2Work Program Grant:** Veterans and First Responders Training Initiative, Award ID AWD000000015868. Role: I served as the main instructor for two cybersecurity courses under the VIT program.
- **U.S. National Science Foundation (NSF):** RINGS: Bringing Post-Quantum Cryptography to Large-Scale NextG Systems, Award Number 2147196. Grant Support: Supported one of my post-quantum cryptography and 5G-related research papers.
- **United States Agency for International Development (USAID):** Critical Infrastructure Digitalization and Resilience Program (CIDR), under USAID Cooperative Agreement No. 7200AA21CA00015, and grant No.1004242-S21-35022-00. Role: I served as an instructor under the USAID-supported program, assisting with course delivery, student support, and instructional activities.

## PROFESSIONAL SERVICE AND REVIEWING

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### Journal Reviewer:

- **IEEE network:** 3 manuscripts
- **Science in China:** 2 manuscripts
- **Computer networks:** 5 manuscripts
- **Springer Nature Computer Science:** 1 manuscript
- **Journal of intelligent & fuzzy systems:** 2 manuscripts
- **Arabian journal for science and engineering:** 2 manuscripts

### Program Committee Member:

- Serving as a TPC member for the 51<sup>st</sup> IEEE LCN 2026 Conference.
- Served as a TPC member for the 44<sup>th</sup> IEEE IPCCC 2025 Workshop.

## **CERTIFICATIONS**

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- Digital Forensics Fundamental Skills.
- Digital and Networking Forensics.

## **AWARDS**

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- Best Paper Award at IEEE LANMAN 2026.
- 2024 & 2025 Ph.D. 1st Author Publication Incentive Award Recipients.

## **LANGUAGES**

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- Languages: Bilingual English and Arabic.